Stochastic optimization

Exercise collection

with solutions

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This collection accompanies the lecture slides. Each exercise names the deck it depends on. This is the version *with solutions*; the statements alone are in `exercises.pdf`.

Sources are indicated exercise by exercise; several are adapted from Birge and Louveaux, *Introduction to Stochastic Programming*, 2nd edition, Springer, 2011.

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1 Generalities

Exercise 1.1. *(Deck 03.) Show that the value of the stochastic solution and the expected value of perfect information are always non-negative.*

Solution.

Consider the general problem $\min_{x\in X} \mathbb{E}[z(x,\boldsymbol{\xi})]$, and recall the four quantities

$$\mathrm{RP} = \min_{x\in X} \mathbb{E}[z(x,\boldsymbol{\xi})], \qquad \mathrm{WS} = \mathbb{E}\left[\min_{x\in X} z(x,\boldsymbol{\xi})\right], \qquad \mathrm{EV} = \min_{x\in X} z(x,\overline{\xi}), \quad \overline{\xi} = \mathbb{E}[\boldsymbol{\xi}],$$

and

$$\mathrm{EEV} = \mathbb{E}\left[z\big(\overline{x}(\overline{\xi}),\boldsymbol{\xi}\big)\right], \qquad \overline{x}(\overline{\xi}) \in \arg\min_{x\in X} z(x,\overline{\xi}).$$

A warning on EEV. It is **not** $\mathbb{E}[\mathrm{EV}]$: EV is a deterministic number, so $\mathbb{E}[\mathrm{EV}] = \mathrm{EV}$. EEV is the expected cost of *using* the expected-value decision, the uncertainty being restored afterwards. This is exactly what makes VSS measure something.

$\mathrm{VSS} = \mathrm{EEV} - \mathrm{RP} \geq 0$. Since $\overline{x}(\overline{\xi}) \in X$, it is one particular feasible point of the recourse problem, so

$$\mathrm{RP} = \min_{x \in X} \mathbb{E}[z(x, \boldsymbol{\xi})] \leq \mathbb{E}\left[z\big(\overline{x}(\overline{\xi}), \boldsymbol{\xi}\big)\right] = \mathrm{EEV}.$$

$\mathrm{EVPI} = \mathrm{RP} - \mathrm{WS} \geq 0$. Let $x^*_{\mathrm{RP}} \in \arg\min_{x \in X} \mathbb{E}[z(x, \boldsymbol{\xi})]$. For every realization ξ,

$$z(x^*_{\mathrm{RP}}, \xi) \geq \min_{x \in X} z(x, \xi),$$

and taking expectations, which preserves inequalities,

$$\mathrm{RP} = \mathbb{E}\left[z(x^*_{\mathrm{RP}}, \boldsymbol{\xi})\right] \geq \mathbb{E}\left[\min_{x \in X} z(x, \boldsymbol{\xi})\right] = \mathrm{WS}.$$

Both arguments have the same shape: a single decision that must serve every realization can only be worse than a decision tailored to each one.

2 Two-stage problems with recourse

Exercise 2.1. *(Deck 01 and 03.) Consider the farmer problem of deck 01. Is its recourse* complete*?* Relatively complete*?* Simple*?*

Solution.

Write t_1, t_2, t_3 for the yields (tons per acre) of one scenario, y_1, y_2 for the tons of wheat and corn bought, w_1, w_2 for the tons sold, and w_3, w_4 for the tons of sugar beets sold at the high and at the low price. Given the acreages x, the second stage is

$$\begin{aligned} \min_{y,w}\ & 238y_1 - 170w_1 + 210y_2 - 150w_2 - 36w_3 - 10w_4 \\ \text{s.t. } & t_1x_1 + y_1 - w_1 \geq 200, \qquad t_2x_2 + y_2 - w_2 \geq 240, \\ & w_3 + w_4 \leq t_3x_3, \qquad w_3 \leq 6000, \qquad y, w \geq 0. \end{aligned}$$

Adding slacks $s \geq 0$ and moving x to the right-hand side,

$$y_1 - w_1 - s_1 = 200 - t_1x_1, \quad y_2 - w_2 - s_2 = 240 - t_2x_2, \quad w_3 + w_4 + s_3 = t_3x_3, \quad w_3 + s_4 = 6000,$$

so that, with the columns ordered as $(y_1, y_2, w_1, w_2, w_3, w_4, s_1, s_2, s_3, s_4)$,

$$W = \begin{pmatrix} 1 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Not complete. Rows 3 and 4 meet only columns with non-negative entries, so every $z \in \operatorname{pos} W$ has $z_3 \geq 0$ and $z_4 \geq 0$. Conversely any such z is attained (take $w_3 = 0$, $s_4 = z_4$, $w_4 = z_3$, $s_3 = 0$, and use (y_1, w_1), (y_2, w_2) for the first two rows, which are unrestricted). Hence $\operatorname{pos} W = \mathbb{R} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+ \neq \mathbb{R}^4$.

Relatively complete. We need $h(\xi) - T(\xi)x \in \operatorname{pos} W$ for every $x \in K_1 = \{x \in \mathbb{R}^3_+ \mid x_1 + x_2 + x_3 \leq 500\}$ and every scenario. Its third and fourth components are t_3x_3 and 6000, both non-negative since $x_3 \geq 0$ and the yields are positive. So $K_1 \subseteq K_2$. Concretely, buying all the wheat and corn needed and selling no sugar beets is feasible whatever the seeding decision and the yields — expensive, but feasible.

Not simple. Simple recourse requires $W = \begin{pmatrix} I & -I \end{pmatrix}$. Here W is 4×10 and w_3 appears in two rows, the quota coupling the beet sales to the harvest.

Completeness is read off W alone and fails here on a technicality; relative completeness — the property that matters — holds because every right-hand side the model can produce lands in $\operatorname{pos} W$.

Exercise 2.2. *(Deck 01.) A newsvendor buys each copy at cost $c = 2$, sells it at price $q = 5$, and gets $r = 1$ back for each unsold copy. The daily demand is $\boldsymbol{\xi} \sim U[0, 100]$.*

(a) Having already bought t copies, what is the expected marginal revenue of one more copy? Deduce the optimal order quantity x^*.

(b) Compare x^* with the expected demand, and explain the gap in terms of the cost of a missed sale and of an unsold copy.

Solution.

(a) The extra copy costs c; it is sold at q if the demand exceeds t, and returned at r otherwise:

$$MR(t) = -c + q\,P[\boldsymbol{\xi} > t] + r\,P[\boldsymbol{\xi} \le t] = -c + q\big(1 - F(t)\big) + rF(t).$$

MR is non-increasing in t: buying pays as long as $MR(t) > 0$, and the optimum is where it vanishes. With $F(t) = t/100$ on $[0, 100]$,

$$F(x^*) = \frac{q - c}{q - r} = \frac{3}{4}, \qquad x^* = 75.$$

MR is the derivative of the expected profit $-cx + \mathcal{Q}(x)$ of deck 01, so this is the same optimality condition.

(b) $\mathbb{E}[\boldsymbol{\xi}] = 50 < 75$. A missed sale costs the lost margin $q - c = 3$, an unsold copy only $c - r = 1$. Shortage being three times as expensive as surplus, the vendor orders enough to meet the demand three times out of four:

$$\frac{q - c}{q - r} = \frac{q - c}{(q - c) + (c - r)}.$$

Ordering the expected demand would be optimal only if the two costs were equal, making x^* the median, and the median coincided with the mean, as for a symmetric distribution.

Exercise 2.3. *(Deck 03; Birge and Louveaux, p. 87.) Consider the second-stage program*

$$Q(x, \xi) = \min_y \{y \mid \xi y = 1 - x,\ y \ge 0\},$$

where $\boldsymbol{\xi}$ follows a triangular distribution on $[0, 1]$, i.e. $P[\boldsymbol{\xi} \le u] = u^2$.

(a) Is the recourse fixed? Why?

(b) Express $K_2(\xi)$ for every $\xi \in [0, 1]$.

(c) Express K_2, K_2^P and $\mathcal{Q}$.

Solution.

(a) The recourse is *not* fixed: here $W \equiv \boldsymbol{\xi}$, which is random. Moreover ξ can take the value 0, so the transformation $y = (1-x)/\xi$ is not defined on all of $\Xi = [0,1]$. In terms of the cone,

$$\operatorname{pos} W = \begin{cases} \{0\} & \text{if } \xi = 0, \\ \mathbb{R}_+ & \text{if } \xi \neq 0. \end{cases}$$

(b) Two cases. $\xi \in (0,1]$*:* since $y \geq 0$ and $\xi > 0$, feasibility requires $1 - x \geq 0$, so $K_2(\xi) = \{x \mid x \leq 1\}$, with $Q(x,\xi) = (1-x)/\xi$ attained at $y^* = (1-x)/\xi$. $\xi = 0$*:* no y satisfies $0 \cdot y = 1 - x$ unless $x = 1$, so $K_2(0) = \{1\}$.

(c) Therefore

$$K_2^P = \{x \mid x \leq 1\} \cap \{1\} = \{1\},$$

while, since $P[\boldsymbol{\xi} = 0] = 0$ and the density is $f(\xi) = 2\xi$,

$$\mathcal{Q}(x) = \int_0^1 \frac{1-x}{\xi}\, 2\xi \, d\xi = 2(1-x), \qquad \forall x \leq 1,$$

so $K_2 = \{x \mid x \leq 1\}$ and $K_2^P \subsetneq K_2$.

The difference is that a point leaves K_2^P as soon as it is infeasible for *one* value of ξ, whatever its probability, whereas K_2 ignores infeasibilities occurring with probability zero. Recall that for a program with a *fixed* recourse — not the case here — and $\boldsymbol{\xi}$ with finite second-order moments, we would have had $K_2 = K_2^P$.

Exercise 2.4. *(Deck 03.) Consider the second-stage program*

$$\begin{aligned} \min_{y} \; & 2y_1 + y_2 \\ \textit{s.t.} \; & y_1 - y_2 \leq 2 - \boldsymbol{\xi} x_1, \\ & y_2 \leq x_2, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Find $K_2(\xi)$ and K_2 when (a) $\boldsymbol{\xi} \sim U[0,1]$; (b) $\boldsymbol{\xi} \sim$ Poisson(λ), $\lambda > 0$. Which properties do you expect of K_2? Is the recourse simple or complete? If not, under which conditions would it be relatively complete?

Solution.

The elementary set. $y_2 \geq 0$ and $y_2 \leq x_2$ force $x_2 \geq 0$. The first constraint then reads $y_1 \leq 2 - \xi x_1 + y_2$, and the largest admissible right-hand side is obtained at $y_2 = x_2$; a feasible $y_1 \geq 0$ exists if and only if $2 - \xi x_1 + x_2 \geq 0$. Hence

$$K_2(\xi) = \{x \mid x_2 \geq 0,\ \xi x_1 \leq 2 + x_2\} = \{x \mid x_2 \geq \max(0, \xi x_1 - 2)\}.$$

(a) $\boldsymbol{\xi} \sim U[0,1]$, $\Xi = [0,1]$. The recourse is fixed ($\boldsymbol{\xi}$ enters only through $T(\boldsymbol{\xi})$) and the distribution has finite second-order moments, so $K_2 = K_2^s = K_2^P = \bigcap_{\xi \in \Xi} K_2(\xi)$. For $x_1 \geq 0$ and $\xi \leq 1$, $\xi x_1 \leq x_1$, so $x \in K_2(1) \Rightarrow x \in K_2(\xi)$; for $x_1 < 0$, $\xi x_1 \leq 0 \leq 2 + x_2$ always. In both cases $K_2(1) \subseteq K_2(\xi)$ for every $\xi \in \Xi$: the intersection is reached at $\xi = 1$ and

$$K_2 = \{x \mid x_2 \geq 0,\ x_1 \leq 2 + x_2\}.$$

(b) $\xi \sim$ Poisson(λ), $\Xi = \{0, 1, 2, \ldots\}$ unbounded. If $x_1 > 0$ then ξx_1 is unbounded above and no x_2 absorbs it; if $x_1 \le 0$ then $\xi x_1 \le 0 \le 2 + x_2$. Hence $K_2 = \{x \mid x_1 \le 0,\ x_2 \ge 0\}$.

Expected properties. Each $K_2(\xi)$ is an intersection of two half-spaces, hence closed and convex, and so is K_2. This is no accident: with a fixed recourse $x \in K_2(\xi) \iff h(\xi) - T(\xi)x \in \operatorname{pos} W$, and $\operatorname{pos} W$ is a closed convex cone. With a finite support K_2 would moreover be polyhedral; here Ξ is infinite, yet both answers happen to be polyhedral.

Type of recourse. Putting the program in standard form with surplus variables $s_1, s_2 \ge 0$,

$$y_1 - y_2 + s_1 = 2 - \xi x_1, \qquad y_2 + s_2 = x_2, \qquad W = \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

The recourse is *not simple* (W is not $\begin{pmatrix} I & -I \end{pmatrix}$), and *not complete*: the second row involves only non-negative coefficients, so $z_2 < 0$ is unreachable and $\operatorname{pos} W = \mathbb{R} \times \mathbb{R}_+ \neq \mathbb{R}^2$. It is *relatively complete* exactly when $K_1 \subseteq K_2$, i.e. when the first stage already enforces $x_2 \ge 0$ together with $x_1 \le 2 + x_2$ in case (a), or $x_1 \le 0$ in case (b). If the first stage merely requires $x \ge 0$, then

$$\text{(a)} \quad K_1 \cap K_2 \subseteq \{x \ge 0 \mid x_1 \le 2 + x_2\}, \qquad \text{(b)} \quad K_1 \cap K_2 \subseteq \{x \mid x_1 = 0,\ x_2 \ge 0\}.$$

Relative completeness is plausible in the first case and very restrictive in the second: an unbounded support is far less forgiving, since *every* realization has to be survived.

Exercise 2.5. *(Deck 03; Birge and Louveaux, p. 102.) Consider the second-stage program*

$$\begin{aligned} \min_{y} \ & 2y_1 + y_2 \\ \textit{s.t.} \ & y_1 + y_2 \ge 1 - x_1, \\ & y_1 \ge \boldsymbol{\xi} - x_1 - x_2, \\ & y_1, y_2 \ge 0. \end{aligned}$$

Show that this program has a complete recourse when $\boldsymbol{\xi}$ has a finite expectation.

Solution.

A direct argument. Completeness asks that for every right-hand side (z_1, z_2) there exist $y_1, y_2 \ge 0$ with $y_1 + y_2 \ge z_1$ and $y_1 \ge z_2$. Taking $y_1 = \max\{0, z_2\}$ satisfies the second, and $y_2 = \max\{0, z_1 - y_1\}$ the first; both are non-negative.

Through $\operatorname{pos} W$. The standard characterisation requires $Wy = z$ with $y \ge 0$, and is stated for equalities: showing that $Wy \ge z$ is solvable is *not* the same statement. So bring the program to the standard form first, with surplus variables $s \ge 0$:

$$y_1 + y_2 - s_1 = 1 - x_1, \qquad y_1 - s_2 = \xi - x_1 - x_2, \qquad W = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}.$$

The second row leaves y_1 free to absorb the sign of z_2:

$$\begin{cases} y_1 = z_2,\ s_2 = 0 & \text{if } z_2 \ge 0, \\ y_1 = 0,\ s_2 = -z_2 & \text{otherwise.} \end{cases}$$

The first row then reads $y_2 - s_1 = z_1 - y_1$. If $z_2 \ge 0$, then $y_2 - s_1 = z_1 - z_2$ and we take

$$\begin{cases} y_2 = z_1 - z_2,\ s_1 = 0 & \text{if } z_1 - z_2 \ge 0, \\ y_2 = 0,\ s_1 = z_2 - z_1 & \text{otherwise;} \end{cases}$$

if $z_2 < 0$, then $y_1 = 0$, $y_2 - s_1 = z_1$, and it suffices to choose

$$\begin{cases} y_2 = z_1,\ s_1 = 0 & \text{if } z_1 \geq 0, \\ y_2 = 0,\ s_1 = -z_1 & \text{otherwise.} \end{cases}$$

Every $z \in \mathbb{R}^2$ is reached, so $\operatorname{pos} W = \mathbb{R}^2$ and the recourse is complete.

The hypothesis on $\mathbb{E}[\boldsymbol{\xi}]$ played no role, and could not have: completeness involves only W, which is fixed. The statement is copied from Birge and Louveaux, where the hypothesis is superfluous at this point. In particular it is *not* what gives $K_2 = \mathbb{R}^2$ — completeness alone does.

What a finite expectation buys is the *finiteness* of the recourse. Deck 03 uses exactly this W with $P[\boldsymbol{\xi} = 2^n] = 2^{-(n+1)}$, $n = 0, 1, 2, \ldots$, for which $\mathbb{E}[\boldsymbol{\xi}] = +\infty$, to exhibit a problem where the recourse is complete — hence $K_2 = \mathbb{R}^2$ — while $\mathcal{Q}(x) = +\infty$ everywhere, so $K_2^s = \emptyset$. Complete recourse and finite recourse are two different things.

Exercise 2.6. *(Deck 03.) A company must order a quantity x of a product to meet a demand d that is unknown at the time of ordering. The unit ordering cost is $c > 0$. If $d > x$, a penalty $b \geq 0$ is incurred per unit of unmet demand; if $d < x$, a holding cost h is incurred per surplus unit.*

(a) Formulate the problem as a two-stage stochastic program.

(b) Is the recourse simple, complete, relatively complete?

(c) What can be expected of the first-stage solution when $b < c$?

Solution.

(a) The first stage is the order; the second stage settles the discrepancy between x and the realized demand:

$$\min_{x \geq 0}\ cx + \mathbb{E}_{\boldsymbol{d}}[Q(x, \boldsymbol{d})], \qquad Q(x, d) = \min_{y^+, y^- \geq 0} \{by^+ + hy^- \mid y^+ - y^- = d - x\}.$$

(b) Here $W = \begin{pmatrix} 1 & -1 \end{pmatrix}$, so the recourse is *simple*, hence also complete and relatively complete. The finiteness condition of the simple-recourse theorem reads $b + h \geq 0$, satisfied as soon as both costs are non-negative.

(c) With $b < c$, an extra ordered unit costs more than the penalty it avoids. Formally, the objective $cx + \mathbb{E}[b(\boldsymbol{d} - x)^+ + h(x - \boldsymbol{d})^+]$ has right derivative at x equal to $c - b\,P[\boldsymbol{d} > x] + h\,P[\boldsymbol{d} < x] \geq c - b > 0$, so the objective is strictly increasing and $x^* = 0$: the company orders nothing and pays the penalty.

Exercise 2.7. *(Deck 03.) A good X must be produced before its demand is known. Each unit produced costs \$2. Demand is discrete, taking the values D_s with probabilities p_s, $s = 1, \ldots, S$. Demand can also be met by buying from an external supplier at \$3 per unit, once it has been observed. How much should be produced now? Formulate the problem as a two-stage stochastic program minimizing the expected production cost.*

Solution.

Let $x \geq 0$ be the number of units produced at the first stage, and $y_s \geq 0$ the number bought from the external supplier in scenario s. Demand is met when $x + y_s \geq D_s$, so

$$\begin{aligned} \min\ & 2x + \sum_{s=1}^{S} p_s\, 3y_s \\ \text{s.t. } & x + y_s \geq D_s, \quad s = 1, \ldots, S, \\ & x \geq 0, \quad y_s \geq 0, \quad s = 1, \ldots, S. \end{aligned}$$

The second stage decomposes scenario by scenario, with $y_s = (D_s - x)^+$, so the model is a simple-recourse one: producing is cheaper than outsourcing, but must be committed to before the demand is known.

Exercise 2.8. *(Deck 03; Higle,* Tutorials in Operations Research*, INFORMS, 2005.) The Dakota Furniture Company makes desks, tables and chairs. Each item consumes lumber and two kinds of skilled labour, finishing and carpentry:*

Resource	*Cost*	*Desk*	*Table*	*Chair*
Lumber (board ft.)	*\$2*	*8*	*6*	*1*
Finishing (hours)	*\$4*	*4*	*2*	*1.5*
Carpentry (hours)	*\$5.2*	*2*	*1.5*	*0.5*
Selling price		*\$60*	*\$40*	*\$10*

Resources must be acquired before the demand is known. Three equally structured demand scenarios are considered, with probabilities 0.3*,* 0.4 *and* 0.3*:*

Scenario	*Desks*	*Tables*	*Chairs*
Low ($p = 0.3$)	*50*	*20*	*200*
Most likely ($p = 0.4$)	*150*	*110*	*225*
High ($p = 0.3$)	*250*	*250*	*500*

Formulate the problem as a two-stage stochastic program maximizing the expected profit. Only the formulation is asked.

Solution.

Treat it as a resource-allocation problem. First stage: x_1 board feet of lumber, x_2 finishing hours, x_3 carpentry hours. Second stage: y_1 desks, y_2 tables, y_3 chairs produced once the demand is known.

$$\min_{x \geq 0} \; 2x_1 + 4x_2 + 5.2x_3 + \mathbb{E}_{\boldsymbol{\xi}}[Q(x, \boldsymbol{\xi})],$$

with

$$\begin{aligned} Q(x, \xi) = \min_{y} \; & -60y_1 - 40y_2 - 10y_3 \\ \text{s.t. } & 8y_1 + 6y_2 + y_3 \leq x_1, \\ & 4y_1 + 2y_2 + 1.5y_3 \leq x_2, \\ & 2y_1 + 1.5y_2 + 0.5y_3 \leq x_3, \\ & y_i \leq d_i(\xi), \; i = 1, 2, 3, \qquad y \geq 0. \end{aligned}$$

Maximizing profit is minimizing its negative, hence the signs in Q. With the three scenarios,

$$\mathbb{E}_{\boldsymbol{\xi}}[Q(x, \boldsymbol{\xi})] = 0.3\, Q(x, \xi_1) + 0.4\, Q(x, \xi_2) + 0.3\, Q(x, \xi_3),$$

where $d(\xi_1) = (50, 20, 200)$, $d(\xi_2) = (150, 110, 225)$ and $d(\xi_3) = (250, 250, 500)$.

Exercise 2.9. *(Deck 03; Birge and Louveaux, p. 142.) Consider*

$$\begin{aligned} \min_{x} \; & x_1 + 4x_2 + \mathbb{E}_{\boldsymbol{\xi}}[Q(x, \boldsymbol{\xi})] \\ \textit{s.t. } & x_1 + x_2 = 1, \quad x \geq 0, \end{aligned}$$

with

$$\begin{aligned} Q(x,\xi) = \min_{y}\ & y_1 + 10y_2 + 10y_3 \\ \textit{s.t. } & y_1 + y_2 - y_3 = \xi + x_1 - 2x_2, \\ & y_1 \le 2, \qquad y \ge 0, \end{aligned}$$

and $\boldsymbol{\xi} \sim U[1,3]$.

(a) Is the recourse fixed, simple, complete, relatively complete? Answer true or false for each, with a brief justification.

(b) For given x and ξ, show, by discussing the value of $\xi + x_1 - 2x_2$, that

$$y^*(x,\xi) = \begin{cases} y_1 = \xi + x_1 - 2x_2,\ y_2 = y_3 = 0 & \text{if } 0 \le \xi + x_1 - 2x_2 \le 2, \\ y_1 = 2,\ y_2 = \xi + x_1 - 2x_2 - 2,\ y_3 = 0 & \text{if } \xi + x_1 - 2x_2 > 2, \\ y_3 = 2x_2 - \xi - x_1,\ y_1 = y_2 = 0 & \text{if } \xi + x_1 - 2x_2 < 0. \end{cases}$$

(c) Which expected property of $Q(x,\xi)$ does $y^*(x,\xi)$ confirm?

(d) Is $\mathcal{Q}(x) = \mathbb{E}_{\boldsymbol{\xi}}[Q(x,\boldsymbol{\xi})]$ differentiable? Why?

(e) Express K_1, $K_2(\xi)$, K_2^P and K_2.

(f) Give the deterministic equivalent (extensive form) of the discretized problem where $\boldsymbol{\xi}$ takes the values 1, 2, 3.

Solution.

(a) Adding a slack y_4 to the second constraint puts the second stage in standard form,

$$y_1 + y_2 - y_3 = \xi + x_1 - 2x_2, \qquad y_1 + y_4 = 2, \qquad W = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Fixed: true, W does not depend on ξ. *Simple*: false, W is not of the form $\begin{pmatrix} I & -I \end{pmatrix}$. *Complete*: false — the second row reads $y_1 + y_4 = z_2$ with $y_1, y_4 \ge 0$, so no $z_2 < 0$ is reachable. *Relatively complete*: true. For any $x \in K_1$ and any ξ, the point

$$y_1 = 0, \quad y_2 = (\xi + x_1 - 2x_2)^+, \quad y_3 = (-\xi - x_1 + 2x_2)^+, \quad y_4 = 2$$

is feasible, the second component of the right-hand side being the constant $2 \ge 0$.

(b) Write $d = \xi + x_1 - 2x_2$. If $d < 0$, the only feasible extreme point is $y_1 = y_2 = 0$, $y_3 = -d$, $y_4 = 2$, hence optimal. If $d \ge 0$, an extreme point has $y_3 = 0$ and $y_1 + y_2 = d$ with $y_1 \le 2$. Since the cost of y_1 is 1 and that of y_2 is 10, y_1 is pushed as far as possible: $y_1 = \min\{d, 2\}$. This splits into $d \le 2$, giving $y_1 = d$, $y_2 = 0$, and $d \ge 2$, giving $y_1 = 2$, $y_2 = d - 2$, as announced.

(c) $Q(\cdot,\xi)$ is convex and piecewise linear in the first-stage decision x: the three regimes above are the three linear pieces, with breakpoints where $d = 0$ and $d = 2$.

(d) Yes. $\boldsymbol{\xi}$ is continuous with an absolutely continuous distribution and finite second-order moments, so $\mathcal{Q}$ is differentiable on the interior of K_2: integrating over ξ smooths away the kinks of each $Q(\cdot,\xi)$.

(e) Directly from the statement, $K_1 = \{x \mid x_1 + x_2 = 1,\ x \ge 0\}$. From (b), the second stage is feasible whatever d, so $K_2(\xi) = \mathbb{R}^2$ for every ξ, and therefore $K_2^P = \bigcap_{\xi \in \Xi} K_2(\xi) = \mathbb{R}^2$ and $K_2 = \mathbb{R}^2$ as well. In particular $K_1 \subseteq K_2$, confirming (a).

(f) With $\xi_1 = 1$, $\xi_2 = 2$, $\xi_3 = 3$ and $p_i = 1/3$,

$$\begin{aligned}
\min\ & x_1 + 4x_2 + \sum_{i=1}^{3} p_i \left(y_{1i} + 10y_{2i} + 10y_{3i}\right) \\
\text{s.t. } & x_1 + x_2 = 1, \\
& y_{1i} + y_{2i} - y_{3i} = \xi_i + x_1 - 2x_2, \quad i = 1,2,3, \\
& y_{1i} \le 2, \quad i = 1,2,3, \\
& x \ge 0, \quad y_{1i}, y_{2i}, y_{3i} \ge 0, \quad i = 1,2,3.
\end{aligned}$$

3 Decomposition methods

Exercise 3.1. *(Deck 04 and LP background.) Apply the dual decomposition method (the L-shaped method with a single scenario) to*

$$\min_{0 \le x \le 5}\ 2x + f(x), \qquad f(x) = \min_{y \ge 0} \left\{ y_1 + 3y_2 \mid y_1 + y_2 = 4 - x,\ \ y_1 + y_3 = 3 \right\}.$$

A demand $4 - x$ must be met exactly by a cheap source y_1 of capacity 3 (y_3 is the slack) and by an expensive source y_2. The master problem is $\min\{2x + \theta \mid 0 \le x \le 5,\ \theta \ge 0,\ \text{cuts}\}$, the constraint $\theta \ge 0$ being valid since $f \ge 0$.

(a) Identify W, q, h and T, and write the dual of the second-stage problem at $\hat{x}$.

(b) Suppose the master returns $\hat{x} = 5$. Using Farkas' lemma, show that the second stage is infeasible by exhibiting a certificate σ, and write the feasibility cut in terms of x.

(c) At the first iteration, $(\hat{x}, \hat{\theta}) = (0, 0)$. Compute $f(0)$ and an optimal dual solution $\hat{\pi}$ (hint: complementary slackness), write the optimality cut, and decide whether the algorithm stops.

(d) Why is the master's value $2\hat{x} + \hat{\theta}$ a lower bound on the optimal value at every iteration?

(e) Carry the method to its end: give the successive iterates, cuts and bounds, and the optimal solution.

Solution.

(a) With $y = (y_1, y_2, y_3)$,

$$W = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad q = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The dual, in asymmetric form (π free in sign), has one constraint per column of W:

$$f(\hat{x}) = \max_{\pi \in \mathbb{R}^2} \left\{ (4 - \hat{x})\pi_1 + 3\pi_2 \mid \pi_1 + \pi_2 \le 1,\ \ \pi_1 \le 3,\ \ \pi_2 \le 0 \right\}.$$

(b) $h - T\hat{x} = (-1, 3)^T$, and $y_1 + y_2 = -1$ is impossible with $y \ge 0$. The vector $\sigma = (-1, 0)^T$ satisfies $W^T\sigma = (-1, -1, 0)^T \le 0$ and $\sigma^T(h - T\hat{x}) = 1 > 0$: it is the second Farkas alternative, with $u = -\sigma$. The feasibility cut is

$$\sigma^T(h - Tx) \le 0 \iff -(4 - x) \le 0 \iff x \le 4.$$

It is valid, since $Wy = h - Tx$ with $y \geq 0$ gives $\sigma^T(h - Tx) = (W^T\sigma)^T y \leq 0$, and it cuts off $\hat{x} = 5$.

(c) To cover 4 with $y_1 \leq 3$: $y_1 = 3$, $y_2 = 1$, $y_3 = 0$, and $f(0) = 6$. By complementary slackness, $y_1 > 0$ forces $\pi_1 + \pi_2 = 1$ and $y_2 > 0$ forces $\pi_1 = 3$, hence $\hat{\pi} = (3, -2)^T$. It is feasible ($\pi_2 \leq 0$), with dual value $4 \cdot 3 + 3 \cdot (-2) = 6 = f(0)$. The optimality cut is

$$\theta \geq \hat{\pi}^T(h - Tx) = 3(4 - x) - 6 = 6 - 3x.$$

Since $f(0) = 6 > \hat{\theta} = 0$, the algorithm does not stop: the cut is added and the master is solved again.

(d) Every cut is valid: it is satisfied by every (x, θ) with $x \in K_1 \cap K_2$ and $\theta \geq f(x)$ — by the argument of (b) for a feasibility cut, and by weak duality, $f(x) \geq \hat{\pi}^T(h - Tx)$ for every dual feasible $\hat{\pi}$, for an optimality cut. The master's feasible set therefore contains that of the reformulated problem $\min\{2x + \theta \mid x \in K_1 \cap K_2,\ \theta \geq f(x)\}$: the master is a relaxation, and its value $2\hat{x} + \hat{\theta}$ cannot exceed the optimal value.

(e) Each iterate also gives an upper bound $2\hat{x} + f(\hat{x})$, the cost of an implementable decision.

- *Iteration 1.* $(\hat{x}, \hat{\theta}) = (0, 0)$, bounds 0 and 6; cut $\theta \geq 6 - 3x$, as in (c).
- *Iteration 2.* The master now has $\theta = \max\{0, 6 - 3x\}$: its objective is $6 - x$ on $[0, 2]$ and $2x$ beyond, so $(\hat{x}, \hat{\theta}) = (2, 0)$ and the lower bound rises to 4. Then $f(2) = 2$, with $y = (2, 0, 1)$; complementary slackness gives $\pi_1 + \pi_2 = 1$ and, since $y_3 > 0$, $\pi_2 = 0$, so $\hat{\pi} = (1, 0)^T$. Upper bound 6; since $f(2) > \hat{\theta}$, add the cut $\theta \geq 4 - x$.
- *Iteration 3.* Now $\theta = \max\{0, 6-3x, 4-x\}$, and $6-3x \geq 4-x$ exactly when $x \leq 1$: the objective is $6-x$ on $[0, 1]$, $x+4$ on $[1, 4]$ and $2x$ on $[4, 5]$, so $(\hat{x}, \hat{\theta}) = (1, 3)$, lower bound 5. Since $f(1) = 3 \leq \hat{\theta}$, the method stops.

The optimal solution is $x^* = 1$, of value 5, and the bounds have closed ($0 \to 4 \to 5$ from below, $6 \to 6 \to 5$ from above). The two cuts are the two linear pieces of $f(x) = \max\{6 - 3x,\ 4 - x\}$ on $[0, 4]$, and $x^* = 1$ is its kink, where both dual vertices are optimal. The feasibility cut of (b) is never generated in this run: the first-stage cost $2x$ keeps $\hat{x}$ away from the infeasible region $x > 4$.

Exercise 3.2. *(Deck 04.) Consider*

$$\begin{aligned}
&\min\ 100x_1 + 150x_2 + \mathbb{E}_{\boldsymbol{\xi}}[q_1 y_1 + q_2 y_2] \\
&\textit{s.t.}\ x_1 + x_2 \leq 120, \\
&\quad 6y_1 + 10y_2 \leq 60x_1, \qquad 8y_1 + 5y_2 \leq 80x_2, \\
&\quad y_1 \leq d_1, \qquad y_2 \leq d_2, \\
&\quad x_1 \geq 40,\ x_2 \geq 20, \qquad y_1, y_2 \geq 0,
\end{aligned}$$

under two scenarios $\boldsymbol{\xi} = (d_1, d_2, q_1, q_2)$*:* $\xi_1 = (500, 100, -24, -28)$ *with probability* $p_1 = 0.4$*, and* $\xi_2 = (300, 300, -28, -32)$ *with probability* $p_2 = 0.6$*.*

Run the first iteration of the single-cut L-shaped method, showing that the second-stage solutions are respectively $y^* = (137.5, 100)$ *and* $y^* = (80, 192)$*. To obtain the dual solutions, use complementary slackness: a dual variable can be non-zero only if the associated primal constraint is active.*

Solution.

Master problem. With no cut yet, $\min\{100x_1 + 150x_2 \mid x_1 + x_2 \leq 120,\ x_1 \geq 40,\ x_2 \geq 20\}$, whose optimal solution is immediately $x^1 = (40, 20)$, so $60x_1 = 2400$ and $80x_2 = 1600$.

Since $x_1, x_2 \geq 0$ makes the second stage always feasible, the recourse is relatively complete and no feasibility cut is needed.

First scenario. $\min -24y_1 - 28y_2$ s.t. $6y_1 + 10y_2 \le 2400$, $8y_1 + 5y_2 \le 1600$, $y_1 \le 500$, $y_2 \le 100$, $y \ge 0$. We cannot have $y_1 = 500$, as the second constraint would be violated. Setting $y_2 = 100$,

$$y_1 = \min\left\{\frac{2400-1000}{6},\ \frac{1600-500}{8}\right\} = \min\left\{\frac{1400}{6},\ \frac{1100}{8}\right\} = \min\{233.33,\ 137.5\} = 137.5.$$

The alternative, making the first two constraints active, solves $3y_1 + 5y_2 = 1200$ and $8y_1 + 5y_2 = 1600$, giving $y_1 = 80$, $y_2 = 192$, which violates $y_2 \le 100$. Hence $y^* = (137.5, 100)$, with objective -6100.

Active constraints: the second and $y_2 \le 100$. Complementary slackness gives $\pi_1^* = \pi_3^* = 0$, and both $y_1, y_2 > 0$ make the dual constraints tight:

$$8\pi_2^* = -24 \ \Rightarrow\ \pi_2^* = -3, \qquad 5\pi_2^* + \pi_4^* = -28 \ \Rightarrow\ \pi_4^* = -28 + 15 = -13.$$

Second scenario. $\min -28y_1 - 32y_2$, same first two constraints, $y_1 \le 300$, $y_2 \le 300$. Taking $y_1 = 300$ or $y_2 = 300$ violates one of the first two constraints, so both are active:

$$3y_1 + 5y_2 = 1200, \qquad 8y_1 + 5y_2 = 1600 \implies y^* = (80, 192),$$

with objective -8384. Then $\pi_3^* = \pi_4^* = 0$ and

$$6\pi_1^* + 8\pi_2^* = -28, \qquad 10\pi_1^* + 5\pi_2^* = -32.$$

Multiplying the first by 5 and the second by 8 and subtracting, $50\pi_1^* = -116$, so $\pi_1^* = -2.32$; then $5\pi_2^* = -32 + 23.2 = -8.8$, so $\pi_2^* = -1.76$.

Optimality cut. Writing the second-stage constraints as $Wy \le h(\xi) - T(\xi)x$,

$$h(\xi_1) = \begin{pmatrix} 0 \\ 0 \\ 500 \\ 100 \end{pmatrix}, \quad h(\xi_2) = \begin{pmatrix} 0 \\ 0 \\ 300 \\ 300 \end{pmatrix}, \quad T(\xi_1) = T(\xi_2) = T = \begin{pmatrix} -60 & 0 \\ 0 & -80 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Hence

$$e_1 = 0.4\,\pi_1^{*T} h(\xi_1) + 0.6\,\pi_2^{*T} h(\xi_2) = 0.4 \times (-1300) + 0.6 \times 0 = -520,$$

and

$$E_1 = 0.4\,\pi_1^{*T} T + 0.6\,\pi_2^{*T} T = 0.4\begin{pmatrix} 0 & 240 \end{pmatrix} + 0.6\begin{pmatrix} 139.2 & 140.8 \end{pmatrix} = \begin{pmatrix} 83.52 & 180.48 \end{pmatrix}.$$

The cut $E_1 x + \theta \ge e_1$ reads

$$\theta \ge -520 - 83.52\,x_1 - 180.48\,x_2,$$

and one checks that it is tight at $x^1 = (40, 20)$: $-520 - 3340.8 - 3609.6 = -7470.4 = 0.4(-6100) + 0.6(-8384) = \mathcal{Q}(x^1)$, as an optimality cut must be at the point where it was generated.

The master problem for the next iteration is therefore

$$\min\{100x_1 + 150x_2 + \theta \mid x_1 + x_2 \le 120,\ x_1 \ge 40,\ x_2 \ge 20,\ \theta \ge -520 - 83.52x_1 - 180.48x_2\}.$$

Exercise 3.3. *(Deck 04; Birge and Louveaux, p. 174.) Consider the problem*

$$\begin{aligned} \min_{x \ge 0}\ & 3x_1 + 2x_2 + \mathbb{E}_{\boldsymbol{\xi}}\left[\min_{y \ge 0}\ -15y_1 - 12y_2\right] \\ \text{s.t. }\ & 3y_1 + 2y_2 \le x_1, \qquad 2y_1 + 5y_2 \le x_2, \\ & 0.8\,\xi_1 \le y_1 \le \xi_1, \qquad 0.8\,\xi_2 \le y_2 \le \xi_2, \end{aligned}$$

where $\boldsymbol{\xi}$ takes the values $\xi^{(1)} = (4, 6)$, $\xi^{(2)} = (4, 8)$, $\xi^{(3)} = (6, 6)$ and $\xi^{(4)} = (6, 8)$, each with probability $1/4$. Started from $x^1 = (0, 0)$, the L-shaped method needs three feasibility cuts, $3x_1 + x_2 \ge 123.2$, then $x_2 \ge 22.4$ and $x_2 \ge 41.6$, to reach a point of K_2, $x^4 = (27.2, 41.6)$.

(a) Show that every $x \in K_2$ satisfies $x_1 \geq 27.2$ and $x_2 \geq 41.6$.

(b) These two constraints are added to the first stage from the start. Can this exclude the optimal solution? Show that the recourse becomes relatively complete.

(c) Is the recourse then complete?

(d) The method, run without these two constraints, never generates $x_1 \geq 27.2$. Why does it still reach a point of K_2, and why does it need three cuts where two constraints suffice?

Solution.

(a) Every scenario must be feasible, in particular $\xi^{(4)} = (6, 8)$, which forces $y_1 \geq 4.8$ and $y_2 \geq 6.4$. Then

$$x_1 \geq 3y_1 + 2y_2 \geq 14.4 + 12.8 = 27.2, \qquad x_2 \geq 2y_1 + 5y_2 \geq 9.6 + 32 = 41.6.$$

These are *induced* constraints: they hold on K_2, although they appear nowhere in the first stage.

(b) No: by (a), they are satisfied by every point of K_2, the optimal solution included, so they exclude nothing useful; the optimal solution, $x^* = (27.2, 41.6)$, even makes both active. This is a reformulation, not an approximation.

Conversely, let $x_1 \geq 27.2$ and $x_2 \geq 41.6$, and take $y = 0.8\,\xi^{(s)}$ in scenario s. The bounds on y hold, and since $\xi^{(4)}$ is the largest scenario componentwise,

$$3y_1 + 2y_2 \leq 3(4.8) + 2(6.4) = 27.2 \leq x_1, \qquad 2y_1 + 5y_2 \leq 2(4.8) + 5(6.4) = 41.6 \leq x_2.$$

Every scenario is feasible, hence $K_2 = \{x \mid x_1 \geq 27.2,\ x_2 \geq 41.6\}$ exactly. With the new first-stage set $K_1 = \{x \geq 0 \mid x_1 \geq 27.2,\ x_2 \geq 41.6\}$, $K_1 \subseteq K_2$: the recourse is relatively complete, and the method only generates optimality cuts.

(c) No. Complete recourse asks the second stage to be feasible for *every* x, whatever the first-stage constraints: it is a property of the second stage alone, which adding constraints to the first stage cannot change. Here $x = (0, 0)$ is still infeasible in every scenario, since the lower bounds on y require some capacity. Relatively complete recourse only asks for feasibility on K_1, and that is what the new constraints achieve.

(d) A feasibility cut comes from the current iterate and from the dual solution returned by the solver for one infeasible scenario. It is valid on K_2 and cuts off the iterate, but nothing makes it describe K_2 as a whole. Here the first cut, $3x_1 + x_2 \geq 123.2$, is 3 times $x_1 \geq 27.2$ plus $x_2 \geq 41.6$: it cuts off $(0, 0)$, but the master moves to $x^2 = (41.067, 0)$, still outside K_2. The second cut, $x_2 \geq 22.4$, is even weaker than the induced constraint $x_2 \geq 41.6$, and leads to $x^3 = (33.6, 22.4)$; only the third gives $x^4 = (27.2, 41.6)$.

The first and third cuts meet at $(27.2, 41.6)$, the corner of K_2, and the first-stage cost $3x_1 + 2x_2$ brings the master there. The method stops generating feasibility cuts as soon as the iterate lies in K_2, not when the master's feasible set equals $K_1 \cap K_2$. After the three cuts, the point $(20, 70)$ satisfies all of them, yet $x_1 < 27.2$: the cuts describe K_2 only where the iterates went, as optimality cuts describe $\mathcal{Q}$ only near the iterates. Exploiting the structure of the problem before running the method, here a scenario that dominates all the others componentwise, saves these iterations.

Exercise 3.4. *(Deck 04.) True or false? Justify your answer.*

(a) If two solvers, applied to the same phase-1 problem at the same point x^ν, return two different feasibility cuts, one of them is wrong.

(b) Like a feasibility cut, an optimality cut removes first-stage decisions x from the master problem.

Solution.

(a) False. The dual of the phase-1 problem may have several optimal solutions, one per optimal basis. Each of them satisfies $W^T\sigma \le 0$ and $\sigma^T(h - Tx^\nu) = w^* > 0$, so each cut $\sigma^T(h - Tx) \le 0$ is valid on K_2, by the argument of Exercise 3.1(b), and excludes x^ν. Which cut is returned depends on the optimal basis the solver ends on, not on an error.

In the example of Exercise 3.3, at $x^1 = (0, 0)$ and for the scenario $\xi^{(4)}$, GLPK returns $x_1 + x_2 \ge 11.2$, while CPLEX or Gurobi return $3x_1 + x_2 \ge 123.2$, the cut reported by Birge and Louveaux. Both are correct: both follow from the induced constraints $x_1 \ge 27.2$ and $x_2 \ge 41.6$, and both exclude $(0, 0)$. They are not equally useful, though: on $x \ge 0$ the second implies the first. The master then moves to $(0, 11.2)$ with the first and to $(41.067, 0)$ with the second: the next iterates, and the number of iterations, differ.

(b) False. An optimality cut $\theta \ge \alpha + \beta^T x$ involves the pair (x, θ): for every x, a large enough θ satisfies it. It removes (x^ν, θ^ν), where θ^ν underestimated $\mathcal{Q}(x^\nu)$, not x^ν itself. It makes x^ν less attractive by raising the model there, and x^ν may even be optimal and come back at the next iteration, with $\theta = \mathcal{Q}(x^\nu)$. In Exercise 3.1, the first optimality cut, $\theta \ge 6 - 3x$, leaves $x = 0$ in the master, with $\theta \ge 6 = f(0)$, whereas the feasibility cut $x \le 4$ of part (b) removes every $x > 4$. Only feasibility cuts, with no term in θ, restrict the admissible x: they approximate K_2, while optimality cuts approximate the epigraph of $\mathcal{Q}$.

Exercise 3.5. *(Deck 04.) In the regularized decomposition method of Ruszczyński, what is the difference between an* exact *serious step and an* approximate *serious step? Why must the two be distinguished?*

Solution.

Briefly: the difference is whether the step can still be the optimal solution. In both cases the incumbent moves because a better objective value has been found, but after an approximate step we *know* the solution is not optimal, whereas after an exact step it might be.

In more detail, in the regularized method:

- Step 4 is reached when *no* optimality cut has been added. In the standard L-shaped method this would already prove that the master solution is optimal, but here the proximal penalty may have pulled the solution away from the true minimizer, so the conclusion does not follow. The current first-stage point is taken as the new incumbent and the method returns to step 1. The stopping test there checks whether the master returns $x^\nu = a^\nu$: the penalty term then vanishes, and since no new cut was produced, the solution is optimal. Otherwise the penalty is what prevented optimality, and the method continues.

- Step 5 can only be reached when the condition of step 4 fails, i.e. when at least one new optimality cut was added at step 3. A point at which a cut has just been generated cannot be optimal — the cut is violated there. It may still give a better value of the full two-stage objective, in which case it is accepted as the new incumbent, but we already know it is not the answer and must keep going.

4 Multistage problems

Exercise 4.1. *(Decks 06 and 08; Heitsch, Römisch and Strugarek,* Stability of multistage stochastic programs*, SIAM Journal on Optimization 17(2):511–525, 2006.) Consider the stylized multistage program below, seeking an optimal purchasing strategy under uncertain costs $\boldsymbol{\xi}_t$ over the periods $t = 0, \dots, T$. The*

amounts purchased are x_t, and a prescribed amount a must be reached at the horizon T:

$$\begin{aligned} &\min_x \mathbb{E}\left[\sum_{t=0}^{T} \boldsymbol{\xi}_t x_t\right], \\ &\text{s.t. } s_t - s_{t-1} = x_t, \quad t = 0, \dots, T, \\ &\qquad s_{-1} = 0, \quad s_T = a, \quad x_t \geq 0, \quad s_t \geq 0, \end{aligned}$$

where s_t is a state variable holding the amount accumulated at time t.

Take three stages ($t = 0, 1, 2$), a binary tree, and $\xi_0 = 10$; the cost goes up by 1 with probability 0.6 and down by 1 with probability 0.4.

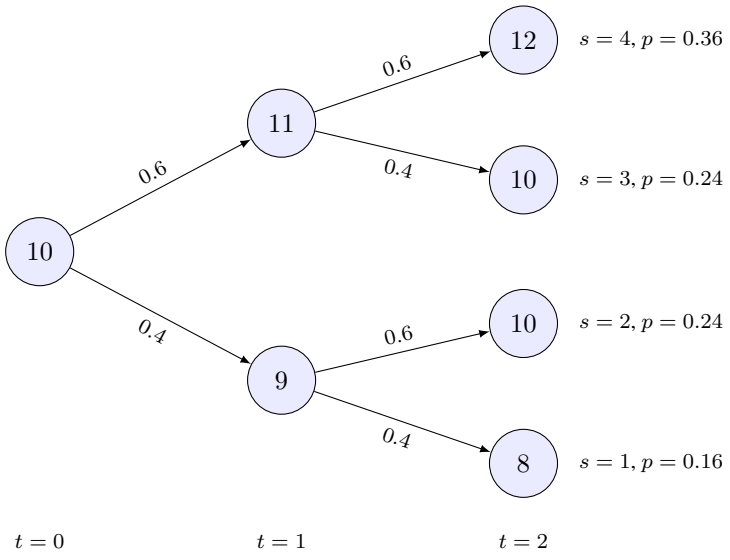
(a) Draw the scenario tree.

(b) Write the problem (i) in extensive form, (ii) using a scenario decomposition.

(c) In which of the two formulations must non-anticipativity constraints be written explicitly?

Solution.

(a) Four scenarios, of probabilities 0.16, 0.24, 0.24 and 0.36:



(b)(i) Extensive form. Attach one decision to each *node* of the tree: x_0 at the root, $x_{1,1}$, $x_{1,2}$ at stage 1, and $x_{2,1}, \dots, x_{2,4}$ at stage 2, with the states s indexed likewise. Writing the cost of a node times the probability of reaching it,

$$\begin{aligned} &\min\ 10x_0 + 0.4\,(9x_{1,1} + 0.4 \cdot 8x_{2,1} + 0.6 \cdot 10x_{2,2}) + 0.6\,(11x_{1,2} + 0.4 \cdot 10x_{2,3} + 0.6 \cdot 12x_{2,4}) \\ &\text{s.t. } s_0 = x_0, \\ &\qquad s_{1,1} = s_0 + x_{1,1}, \qquad s_{1,2} = s_0 + x_{1,2}, \\ &\qquad s_{2,1} = s_{1,1} + x_{2,1}, \quad s_{2,2} = s_{1,1} + x_{2,2}, \quad s_{2,3} = s_{1,2} + x_{2,3}, \quad s_{2,4} = s_{1,2} + x_{2,4}, \\ &\qquad s_{2,j} = a, \quad j = 1, \dots, 4, \\ &\qquad x \geq 0, \quad s \geq 0, \end{aligned}$$

where node $(1,1)$ carries $\xi_1 = 9$ and node $(1,2)$ carries $\xi_1 = 11$.

(b)(ii) Scenario decomposition. Attach instead one full decision path to each *scenario*: $x_t^{(s)}$ and $s_t^{(s)}$, $t = 0,1,2$, $s = 1,\dots,4$. With $p = (0.16, 0.24, 0.24, 0.36)$ and the cost paths $(10,9,8)$, $(10,9,10)$, $(10,11,10)$, $(10,11,12)$,

$$
\begin{aligned}
\min\ & 0.16\left(10x_0^{(1)} + 9x_1^{(1)} + 8x_2^{(1)}\right) + 0.24\left(10x_0^{(2)} + 9x_1^{(2)} + 10x_2^{(2)}\right) \\
& + 0.24\left(10x_0^{(3)} + 11x_1^{(3)} + 10x_2^{(3)}\right) + 0.36\left(10x_0^{(4)} + 11x_1^{(4)} + 12x_2^{(4)}\right) \\
\text{s.t. } & s_0^{(s)} = x_0^{(s)}, \quad s_t^{(s)} = s_{t-1}^{(s)} + x_t^{(s)}, \quad t = 1,2, \quad s = 1,\dots,4, \\
& s_2^{(s)} = a, \quad x_t^{(s)} \geq 0, \quad s_t^{(s)} \geq 0, \\
& x_0^{(1)} = x_0^{(2)} = x_0^{(3)} = x_0^{(4)}, && \text{(non-anticipativity, } t = 0) \\
& x_1^{(1)} = x_1^{(2)}, \qquad x_1^{(3)} = x_1^{(4)}. && \text{(non-anticipativity, } t = 1)
\end{aligned}
$$

(c) Only in the scenario decomposition. The extensive form attaches one variable per *node*, so two scenarios that are indistinguishable at time t automatically share the same decision: non-anticipativity is built into the indexing. The scenario decomposition duplicates the decisions along each path, which is what makes the problem separable by scenario, and the coupling must then be restored explicitly. Those are precisely the constraints that deck 08 dualizes, with multipliers π, and that progressive hedging enforces iteratively.

Note the pattern: at $t = 2$ the four scenarios are all distinguishable, so no constraint is needed; at $t = 1$ they form two groups; at $t = 0$ a single one.

5 Random number generation

Exercise 5.1. *(Deck 02.) A linear congruential generator produces* $x_i = (ax_{i-1} + c) \bmod m$.

(a) For $m = 16$ *and* $c = 1$*, use the period theorem to find every* a *giving a full period.*

(b) The "standard minimal" generator of Park and Miller is $x_i = 16807\, x_{i-1} \bmod (2^{31} - 1)$*. Why can its period not be* m*? What is the largest it could be, and why is that value actually reached here?*

(c) For $m = 16$*, which of the pairs* $(a,c) = (3,4)$*,* $(5,3)$*,* $(7,3)$*,* $(4,1)$ *give a full period? For* $(4,1)$*, write the sequence from* $x_0 = 0$*: what happens, and why?*

Solution.

(a) The three conditions are: $\gcd(m,c) = 1$; every prime dividing m divides $a - 1$; and if $4 \mid m$ then $4 \mid a - 1$.

Here $m = 16 = 2^4$ and $c = 1$, so $\gcd(16,1) = 1$ and the first holds. The only prime dividing 16 is 2, so the second requires a odd. Since $4 \mid 16$, the third requires $4 \mid a - 1$, which is strictly stronger. Hence

$$\text{full period} \iff a \equiv 1 \pmod 4, \quad \text{i.e. } a \in \{1,5,9,13\} \text{ modulo } 16.$$

(b) Here $c = 0$, so $x = 0$ is a fixed point of the recurrence: once the state hits 0 it stays there, and 0 cannot belong to a cycle of length m. The period is therefore at most $m - 1$, and that bound is reachable only when m is prime — which is the case, $2^{31} - 1$ being a Mersenne prime.

Reaching $m - 1 = 2^{31} - 2$ requires the multiplier to be a primitive root modulo m, i.e. to generate the whole multiplicative group $(\mathbb{Z}/m\mathbb{Z})^*$. That is exactly what Park and Miller checked for $a = 16807 = 7^5$.

Note the contrast with $m = 2^k$, where the period of a multiplicative generator collapses to $m/4$: the modulus being prime is what makes the full multiplicative group available.

(c) For $m = 16$ the three conditions read: c odd, and $a \equiv 1 \pmod 4$. Only $(5, 3)$ satisfies them, and indeed, from $x_0 = 0$,

$$0,\ 3,\ 2,\ 13,\ 4,\ 7,\ 6,\ 1,\ 8,\ 11,\ 10,\ 5,\ 12,\ 15,\ 14,\ 9,\ 0,$$

visits all sixteen values. The pair $(3, 4)$ fails the first condition ($\gcd(4, 16) = 4$), and $(7, 3)$ the third ($4 \nmid 6$): both have period 4, e.g. $1, 7, 9, 15, 1$ and $0, 3, 8, 11, 0$.

The pair $(4, 1)$ fails the second condition ($a - 1 = 3$ is odd), and the sequence from 0 is

$$0,\ 1,\ 5,\ 5,\ 5,\ \ldots$$

5 is a fixed point, since $4 \cdot 5 + 1 = 21 \equiv 5$, and the seed is never visited again. The reason is that $a = 4$ is not invertible modulo 16: f is not a bijection, several states lead to 5 (namely 1, 5, 9 and 13), and the sequence only becomes periodic after a transient — the $\tau > 0$ of the definition $s_{n+\rho} = s_n$ for $n \geq \tau$. With a odd, f is a bijection of $\mathbb{Z}_{16}$ and every sequence is purely periodic ($\tau = 0$).

Exercise 5.2. *(Deck 02.) Consider the linear feedback shift register over $\mathbb{F}_2$ with $k = 4$ and $x_n = x_{n-1} \oplus x_{n-4}$, seeded at* 0001.

(a) Why is the all-zero state excluded, and why can the period not exceed $2^k - 1$?

(b) A Tausworthe generator reads the output stream w bits at a time, spaced ν apart. The slides take $w = 3$, $\nu = 3$ and observe only five distinct values instead of fifteen. Explain, and find every $\nu \in \{1, \ldots, 14\}$ giving the maximal output period.

Solution.

(a) The recurrence is linear over $\mathbb{F}_2$, so the all-zero state maps to itself: it is absorbing, and a register that reaches it never leaves. It must therefore be excluded from any cycle. The state is k bits, so there are 2^k states in all and at most $2^k - 1 = 15$ usable ones. This register attains that bound — its characteristic polynomial $z^4 - z^3 - 1$ is primitive over $\mathbb{F}_2$ — so all fifteen non-zero states are visited.

(b) Taking every ν-th bit of a sequence of period $\rho = 2^k - 1$ produces a sequence of period

$$\frac{\rho}{\gcd(\nu, \rho)}.$$

With $\nu = 3$ and $\rho = 15$, $\gcd(3, 15) = 3$ and the output period falls to $15/3 = 5$: the five values $7/8$, $5/8$, $3/8$, $1/8$, 0 of the slides, repeating forever.

The output period is maximal exactly when $\gcd(\nu, 15) = 1$. Excluding the multiples of 3 and of 5,

$$\nu \in \{1, 2, 4, 7, 8, 11, 13, 14\},$$

that is $\varphi(15) = 8$ admissible spacings out of 14. This is the coprimality condition of the slides, on the smallest example where it bites.

Exercise 5.3. *(Deck 02.) Let X take the values* 1, 2, 3 *with probabilities* 0.2, 0.5, 0.3.

(a) Write F_X and its generalized inverse, and check that $F_X^{-1}(U)$, $U \sim U[0,1]$, has the right distribution.

(b) Why can the ordinary inverse not be used here?

(c) Inversion is called the "gold standard" in the slides. Which property makes common random numbers and antithetic variates reliable with it?

(d) A demand Y is zero with probability $1/2$ and, otherwise, uniform on $[0,2]$. Write F_Y and its generalized inverse, and give the values generated for $U = 1/4$ and $U = 3/4$. Which values of U produce $Y = 0$, and why is that correct?

Solution.

(a) F_X is the step function $F_X(x) = 0$ for $x < 1$, 0.2 on $[1,2)$, 0.7 on $[2,3)$ and 1 for $x \geq 3$. Its generalized inverse $F_X^{-1}(u) = \inf\{x \mid F_X(x) \geq u\}$ is

$$F_X^{-1}(u) = \begin{cases} 1 & 0 < u \leq 0.2, \\ 2 & 0.2 < u \leq 0.7, \\ 3 & 0.7 < u \leq 1. \end{cases}$$

Then $P[F_X^{-1}(U) = 1] = P[U \leq 0.2] = 0.2$, $P[F_X^{-1}(U) = 2] = P[0.2 < U \leq 0.7] = 0.5$ and $P[F_X^{-1}(U) = 3] = 0.3$: the masses are reproduced. The lengths of the three sub-intervals of $[0,1]$ *are* the probabilities, which is the whole mechanism.

(b) F_X is neither continuous nor injective: it jumps at 1, 2, 3 and is constant in between. It has no inverse in the usual sense. The $\inf$ in the generalized inverse picks, among the x with $F_X(x) \geq u$, the leftmost — which is what makes the flat stretches harmless.

(c) Monotonicity. F_X^{-1} is non-decreasing, so a single uniform U drives the output in a monotone way: reusing the same U across two systems induces a correlation of known sign, which is what makes the variance of a difference shrink rather than grow. Two consequences, both in the slides: one uniform per variate, and synchronization preserved. Methods consuming a random number of uniforms — acceptance–rejection, the ziggurat behind `randn()` — lose both.

(d) The distribution is mixed: an atom of mass $1/2$ at 0, and a continuous part of density $1/4$ on $[0,2]$. Hence $F_Y(y) = 0$ for $y < 0$, $F_Y(y) = \frac{1}{2} + \frac{y}{4}$ on $[0,2)$ and $F_Y(y) = 1$ for $y \geq 2$, with a jump of $\frac{1}{2}$ at 0. The same $\inf$ gives

$$F_Y^{-1}(u) = \begin{cases} 0 & 0 < u \leq \frac{1}{2}, \\ 4u - 2 & \frac{1}{2} < u < 1, \end{cases}$$

so $U = 1/4$ gives $Y = 0$ and $U = 3/4$ gives $Y = 1$. Every $U \leq \frac{1}{2}$ produces $Y = 0$: an interval of length $\frac{1}{2}$, which is exactly the mass of the atom. The jump of F_Y becomes an interval of values of U, the continuous part is inverted as usual, and nothing requires F_Y to be invertible. Two tempting mistakes: ignoring the atom and taking $Y = 2U$, which gives $\frac{1}{2}$ and $\frac{3}{2}$; or extending $\frac{1}{2} + \frac{y}{4}$ below 0, which gives -1 for $U = 1/4$, outside the support.

Exercise 5.4. *(Deck 02.) In acceptance–rejection one takes $t \geq f$ with $\int t = a$, so that $r = t/a$ is a density, then repeats: draw $Y \sim r$ and $U \sim U[0,1]$ until $U \leq f(Y)/t(Y)$, and return Y.*

(a) Compute the probability that one iteration is accepted.

(b) Deduce the expected number of iterations, and explain why the slides ask for a close to 1.

(c) Why does this method break common random numbers, even when the accepted value is correct?

Solution.

(a) Conditioning on Y and using $U \perp Y$,

$$P\left[U \le \frac{f(Y)}{t(Y)}\right] = \int r(y)\,\frac{f(y)}{t(y)}\,dy = \int \frac{t(y)}{a}\,\frac{f(y)}{t(y)}\,dy = \frac{1}{a}\int f(y)\,dy = \frac{1}{a},$$

since f is a density. The acceptance probability is $1/a$, and does not depend on the shape of f — only on the area wasted between f and t.

(b) Iterations are i.i.d. Bernoulli trials of success probability $1/a$, so their number is geometric with mean a. Each iteration consumes at least two uniforms (one for Y, one for U), hence about $2a$ per variate. This is why t should hug f: a is the cost multiplier, and $a = 1$ would mean $t = f$, the case where nothing is rejected.

(c) The *number* of uniforms consumed is random. Two runs of a simulation compared under common random numbers then read the same stream at different offsets: after the first rejection, the two runs are no longer looking at the same draws, and the intended pairing is lost. The returned value is still correctly distributed — rejection is exact — but the synchronization on which the variance reduction depends is destroyed. This is the trade-off the slides set against inversion.

Exercise 5.5. *(Deck 02.) Let C be a bivariate copula, so that $H(x, y) = C(F_X(x), F_Y(y))$ is the joint cdf of (X, Y).*

(a) Show that the independence copula $C(u, v) = uv$ satisfies the Fréchet–Hoeffding bounds.

(b) The upper bound is attained by $C(u, v) = \min(u, v)$. Which dependence between X and Y does it describe?

(c) A tempting proof of the upper bound argues $P[A \cap B] \le P[A]\,P[B] \le \min(P[A], P[B])$. Give a counterexample to the first inequality, and state the correct argument.

Solution.

(a) For $u, v \in [0, 1]$, $uv \le u$ and $uv \le v$, so $uv \le \min(u, v)$. For the lower bound, $(1 - u)(1 - v) \ge 0$ expands to $uv \ge u + v - 1$, and $uv \ge 0$; hence $uv \ge \max(u + v - 1, 0)$.

(b) It describes comonotonicity: X and Y are increasing functions of one another. Taking a single $U \sim U[0, 1]$ and setting $X = F_X^{-1}(U)$, $Y = F_Y^{-1}(U)$ gives

$$P[X \le x,\ Y \le y] = P\left[U \le F_X(x),\ U \le F_Y(y)\right] = \min\left(F_X(x), F_Y(y)\right).$$

It is the strongest positive dependence compatible with the given margins — and, incidentally, exactly what inversion driven by common random numbers produces.

(c) The inequality $P[A \cap B] \le P[A]P[B]$ is false in general; it holds with equality under independence and fails in the other direction under positive dependence. Take $X = Y$ uniform on $\{0, 1\}$, $A = \{X \le 0\}$, $B = \{Y \le 0\}$: then $A = B$, so

$$P[A \cap B] = \tfrac{1}{2} \qquad \text{while} \qquad P[A]\,P[B] = \tfrac{1}{4}.$$

The upper bound needs no independence, only monotonicity of the measure:

$$\{X \le x\} \cap \{Y \le y\} \subseteq \{X \le x\} \quad \text{and} \quad \subseteq \{Y \le y\},$$

so $H(x, y) \le F_X(x)$ and $H(x, y) \le F_Y(y)$, hence $H(x, y) \le \min(F_X(x), F_Y(y))$.

6 Monte Carlo approximation

Exercise 6.1. *(Decks 03 and 09.) Show that*

$$\mathbb{E}_{\boldsymbol{\xi}}\left[\min_{x\in\mathcal{X}} z(x,\boldsymbol{\xi})\right] \le \min_{x\in\mathcal{X}} \mathbb{E}_{\boldsymbol{\xi}}\left[z(x,\boldsymbol{\xi})\right]$$

for a deterministic feasible set $\mathcal{X}$. How is this result to be interpreted for stochastic programming? Extend it to a Monte Carlo sample average.

Solution.

Assume the expectations are well defined and the minima attained, as the statement implicitly does. There exists $x^* \in \mathcal{X}$ with $\min_{x\in\mathcal{X}} \mathbb{E}_{\boldsymbol{\xi}}[z(x,\boldsymbol{\xi})] = \mathbb{E}_{\boldsymbol{\xi}}[z(x^*,\boldsymbol{\xi})]$. Since $x^* \in \mathcal{X}$,

$$\min_{x\in\mathcal{X}} z(x,\xi) \le z(x^*,\xi) \quad \text{for every } \xi,$$

and taking expectations gives the result.

Without attainment the statement still holds with infima: for every $y \in \mathcal{X}$, $\inf_{x\in\mathcal{X}} z(x,\boldsymbol{\xi}) \le z(y,\boldsymbol{\xi})$, hence $\mathbb{E}_{\boldsymbol{\xi}}[\inf_{x\in\mathcal{X}} z(x,\boldsymbol{\xi})] \le \mathbb{E}_{\boldsymbol{\xi}}[z(y,\boldsymbol{\xi})]$, and taking the infimum over y closes the argument.

Interpretation. This is WS $\le$ RP of deck 03: minimizing under perfect information is optimistic. A decision tailored to each realization can only beat a single decision that must serve them all.

Extension to sampling. Let $z_N(x) = \frac{1}{N}\sum_{n=1}^{N} z(x,\boldsymbol{\xi}_n)$ be the sample average over an i.i.d. sample. The result just proved applies verbatim with the random element taken to be the whole sample $(\boldsymbol{\xi}_1,\ldots,\boldsymbol{\xi}_N)$ and the function z_N in place of z:

$$\mathbb{E}\left[\min_{x\in\mathcal{X}} z_N(x)\right] \;\le\; \min_{x\in\mathcal{X}} \mathbb{E}\left[z_N(x)\right] \;=\; \min_{x\in\mathcal{X}} \mathbb{E}_{\boldsymbol{\xi}}\left[z(x,\boldsymbol{\xi})\right],$$

the last equality because $z_N(x)$ is an unbiased estimator of $\mathbb{E}_{\boldsymbol{\xi}}[z(x,\boldsymbol{\xi})]$ for each fixed x.

Caution. One cannot argue by exchanging the minimum and the sum, writing $\min_x \frac{1}{N}\sum_n z(x,\boldsymbol{\xi}_n) = \frac{1}{N}\sum_n \min_x z(x,\boldsymbol{\xi}_n)$: the minimum of a sum is not the sum of the minima, and the inequality in fact runs the other way,

$$\min_{x\in\mathcal{X}} \frac{1}{N}\sum_{n=1}^{N} z(x,\boldsymbol{\xi}_n) \;\ge\; \frac{1}{N}\sum_{n=1}^{N} \min_{x\in\mathcal{X}} z(x,\boldsymbol{\xi}_n),$$

since a single x must serve all N draws. That route does not close the chain; the one above does.

The conclusion is that the sample-average optimal value is a *downward-biased* estimator of the true optimal value: the optimizer exploits the particular sample it was given. This is the bias that deck 09 analyses, and why an out-of-sample evaluation is needed to obtain an honest upper bound.